

Why 30% of Swiggy Orders Are Delivered Late : A Data Case Study

```
In [115... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
In [116... df = pd.read_csv('Food_Delivery_Dataset.csv')
df.head(5)
```

```
Out[116... 
```

	ID	Delivery_person_ID	Delivery_person_Age	Delivery_person_Ratings	Restaurant_latitude	Restaurant_longitude	Delivery
0	0x4607	INDORES13DEL02	37	4.9	22.745049	75.892471	
1	0xb379	BANGRES18DEL02	34	4.5	12.913041	77.683237	
2	0x5d6d	BANGRES19DEL01	23	4.4	12.914264	77.678400	
3	0x7a6a	COIMBRES13DEL02	38	4.7	11.003669	76.976494	
4	0x70a2	CHENRES12DEL01	32	4.6	12.972793	80.249982	



```
In [117... df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45593 entries, 0 to 45592
Data columns (total 20 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                     45593 non-null  object
1   Delivery_person_ID                    45593 non-null  object
2   Delivery_person_Age                   45593 non-null  object
3   Delivery_person_Ratings               45593 non-null  object
4   Restaurant_latitude                   45593 non-null  float64
5   Restaurant_longitude                  45593 non-null  float64
6   Delivery_location_latitude            45593 non-null  float64
7   Delivery_location_longitude           45593 non-null  float64
8   Order_Date                           45593 non-null  object
9   Time_Orderd                           45593 non-null  object
10  Time_Order_picked                     45593 non-null  object
11  Weatherconditions                     45593 non-null  object
12  Road_traffic_density                  45593 non-null  object
13  Vehicle_condition                     45593 non-null  int64
14  Type_of_order                         45593 non-null  object
15  Type_of_vehicle                       45593 non-null  object
16  multiple_deliveries                   45593 non-null  object
17  Festival                              45593 non-null  object
18  City                                  45593 non-null  object
19  Time_taken(min)                       45593 non-null  object
dtypes: float64(4), int64(1), object(15)
memory usage: 7.0+ MB

```

```
In [118... df.describe().round()
```

Out[118...

	Restaurant_latitude	Restaurant_longitude	Delivery_location_latitude	Delivery_location_longitude	Vehicle_condition
count	45593.0	45593.0	45593.0	45593.0	45593.0
mean	17.0	70.0	17.0	71.0	1.0
std	8.0	23.0	7.0	21.0	1.0
min	-31.0	-88.0	0.0	0.0	0.0
25%	13.0	73.0	13.0	73.0	0.0
50%	19.0	76.0	19.0	76.0	1.0
75%	23.0	78.0	23.0	78.0	2.0
max	31.0	88.0	31.0	89.0	3.0

Handling Missing Values

Even without Null Values Few Columns have NaN as a value

```
In [119... df['Delivery_person_Age'].unique()
```

```
Out[119... array(['37', '34', '23', '38', '32', '22', '33', '35', '36', '21', '24',
      '29', '25', '31', '27', '26', '20', 'NaN ', '28', '39', '30', '15',
      '50'], dtype=object)
```

```
In [120... df['Delivery_person_Ratings'].unique()
```

```
Out[120... array(['4.9', '4.5', '4.4', '4.7', '4.6', '4.8', '4.2', '4.3', '4', '4.1',
      '5', '3.5', 'NaN ', '3.8', '3.9', '3.7', '2.6', '2.5', '3.6',
      '3.1', '2.7', '1', '3.2', '3.3', '6', '3.4', '2.8', '2.9', '3'],
      dtype=object)
```

```
In [121... df['Road_traffic_density'].unique()
```

```
Out[121... array(['High ', 'Jam ', 'Low ', 'Medium ', 'NaN '], dtype=object)
```

```
In [122... df['multiple_deliveries'].unique()
```

```
Out[122... array(['0', '1', '3', 'NaN ', '2'], dtype=object)
```

```
In [123... df['City'].unique()
```

```
Out[123... array(['Urban ', 'Metropolitan ', 'Semi-Urban ', 'NaN '], dtype=object)
```

```
In [124... cols = ['Delivery_person_Age', 'Delivery_person_Ratings', 'Time_Orderd', 'Road_traffic_density', 'multiple_deliveries']
df = df[df[cols].notna().all(axis=1)]
for col in cols:
    df = df[df[col].astype(str).str.strip() != 'NaN']
```

```
In [125... df['Weatherconditions'] = df['Weatherconditions'].replace('conditions NaN', np.nan)
df = df.dropna(subset = 'Weatherconditions')
```

```
In [126... df['Weatherconditions'].unique()
```

```
Out[126... array(['conditions Sunny', 'conditions Stormy', 'conditions Sandstorms',
       'conditions Cloudy', 'conditions Fog', 'conditions Windy'],
       dtype=object)
```

```
In [127... cols = ['Restaurant_latitude', 'Restaurant_longitude', 'Delivery_location_latitude', 'Delivery_location_longitude']
for col in cols:
    df[col] = df[col].replace(0, np.nan)
df = df.dropna(subset = cols)
```

Formatting Inconsistent Values

```
In [128... df['Weatherconditions'].unique()
```

```
Out[128... array(['conditions Sunny', 'conditions Stormy', 'conditions Sandstorms',
       'conditions Cloudy', 'conditions Fog', 'conditions Windy'],
       dtype=object)
```

```
In [129... df['Weatherconditions'] = df['Weatherconditions'].str.replace('conditions ', '', regex=False)
```

```
In [130... df['Time_taken(min)'] = df['Time_taken(min)'].str.replace('(min) ', '', regex=False)
df['Time_taken(min)'] = df['Time_taken(min)'].astype('int64')
```

Changing Datatypes

```
In [131... df['Delivery_person_Age'] = df['Delivery_person_Age'].astype('int64')
df['multiple_deliveries'] = df['multiple_deliveries'].astype('int64')
```

```
In [132... df['Delivery_person_Ratings'] = df['Delivery_person_Ratings'].astype('float64')
```

```
In [133... df['Order_Date'] = df['Order_Date'].astype('datetime64[ns]')
df['Time_Orderd'] = df['Time_Orderd'].astype('datetime64[ns]')
df['Time_Order_picked'] = df['Time_Order_picked'].astype('datetime64[ns]')
```

```
In [134... df['Order_hour'] = df['Time_Orderd'].dt.hour
df['Order_minute'] = df['Time_Orderd'].dt.minute
```

```
In [135... df['Picked_hour'] = df['Time_Order_picked'].dt.hour
df['Picked_minute'] = df['Time_Order_picked'].dt.minute
```

```
In [136... df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 38243 entries, 0 to 45592
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                    38243 non-null  object
1   Delivery_person_ID                   38243 non-null  object
2   Delivery_person_Age                  38243 non-null  int64
3   Delivery_person_Ratings              38243 non-null  float64
4   Restaurant_latitude                  38243 non-null  float64
5   Restaurant_longitude                 38243 non-null  float64
6   Delivery_location_latitude           38243 non-null  float64
7   Delivery_location_longitude          38243 non-null  float64
8   Order_Date                           38243 non-null  datetime64[ns]
9   Time_Orderd                          38243 non-null  datetime64[ns]
10  Time_Order_picked                    38243 non-null  datetime64[ns]
11  Weatherconditions                    38243 non-null  object
12  Road_traffic_density                 38243 non-null  object
13  Vehicle_condition                   38243 non-null  int64
14  Type_of_order                        38243 non-null  object
15  Type_of_vehicle                      38243 non-null  object
16  multiple_deliveries                  38243 non-null  int64
17  Festival                             38243 non-null  object
18  City                                 38243 non-null  object
19  Time_taken(min)                      38243 non-null  int64
20  Order_hour                           38243 non-null  int32
21  Order_minute                         38243 non-null  int32
22  Picked_hour                          38243 non-null  int32
23  Picked_minute                        38243 non-null  int32
dtypes: datetime64[ns](3), float64(5), int32(4), int64(4), object(8)
memory usage: 6.7+ MB
```

```
In [137... # Extracting Late Delivery from Time_Taken
delay_threshold = 30 # minutes
df['Late_Delivery'] = df['Time_taken(min)'] > delay_threshold
```

1. What percentage of orders are Delivered late??

```
In [138... late_percentage = (df['Late_Delivery'].mean()) * 100
print(f'Late Delivery : {late_percentage:.2f}%')
```

Late Delivery : 30.59%

2. What is the Average delay time (in minutes) for late deliveries??

```
In [139... avg_delay = df[df['Late_Delivery']][['Time_taken(min)']].mean()
print(f'Average Delay Time : {avg_delay:.2f} minutes')
```

Average Delay Time : 37.95 minutes

3. How does Delivery Distance affect the probability of late Delivery??

```
In [140... # Calculating Distance using Latitude and Longitude
from geopy.distance import geodesic
def calc_distance(row):
    return geodesic((row['Restaurant_latitude'], row['Restaurant_longitude']),
                    (row['Delivery_location_latitude'], row['Delivery_location_longitude'])).km

df['distance_km'] = df.apply(calc_distance, axis = 1)
```

```
In [141... # Creating Distance bins to calculate late delivery probability
bins = [0,2,5,10,20]
labels = ['0-2 km', '2-5 km', '5-10 km', '10+ km']
df['Distance_Band'] = pd.cut(df['distance_km'], bins=bins, labels=labels)

late_prob = (df.groupby('Distance_Band')['Late_Delivery'].mean() * 100)
print(late_prob)
```

```
Distance_Band
0-2 km      10.361021
2-5 km      13.858175
5-10 km     21.551387
10+ km      46.215711
Name: Late_Delivery, dtype: float64
```

```
In [142... # Plot
plt.figure(figsize=(8,4))
colors = ['#4CAF50', '#FFC107', '#FF9800', '#F44336']
bars = plt.bar(late_prob.index.astype(str), late_prob.values,color=colors)

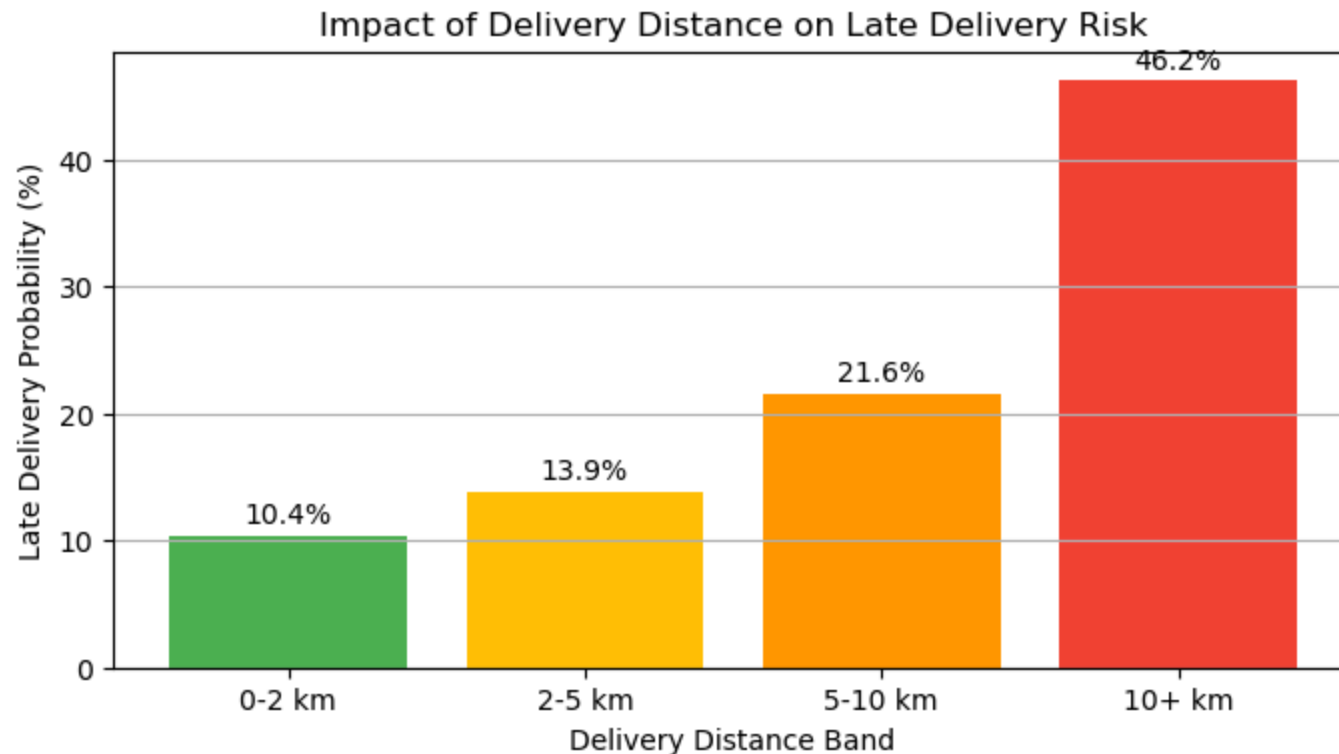
#Add data labels
```

```

for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, height + 1, f'{height:.1f}%', ha='center', fontsize=10)

plt.title('Impact of Delivery Distance on Late Delivery Risk')
plt.xlabel('Delivery Distance Band')
plt.ylabel('Late Delivery Probability (%)')
plt.grid(axis='y')
plt.show()

```



4. During which hours of the day do late deliveries occur most frequently??

```

In [143... # Which hour have most Late orders
late_by_hour = (df[df['Late_Delivery']].groupby('Order_hour').size().sort_values(ascending = False))
print(late_by_hour)

```

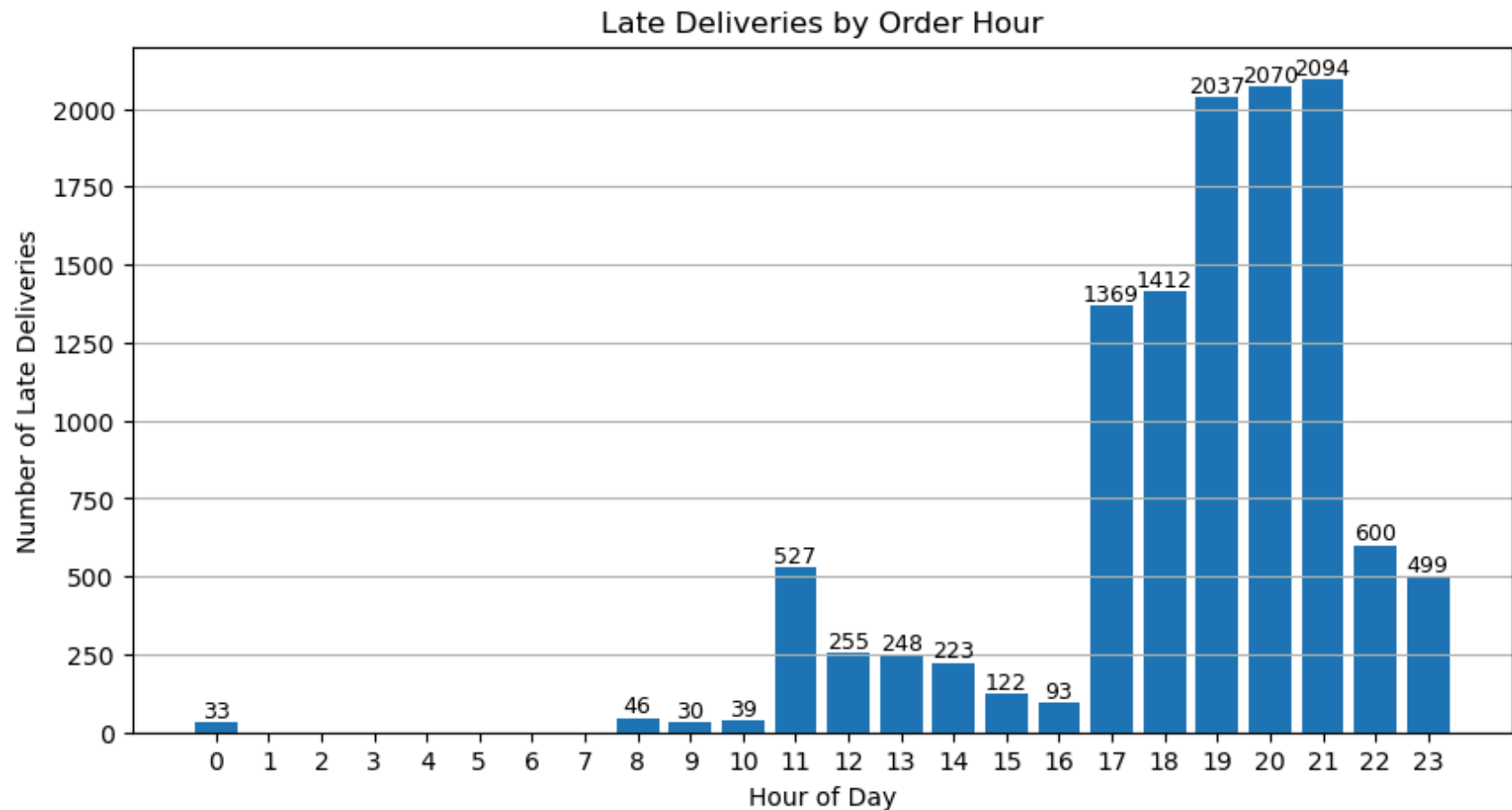


```
Order_hour
21    2094
20    2070
19    2037
18    1412
17    1369
22     600
11     527
23     499
12     255
13     248
14     223
15     122
16      93
 8      46
10      39
 0       33
 9       30
dtype: int64
```

```
In [144... plt.figure(figsize=(10,5))
bars = plt.bar(late_by_hour.index, late_by_hour.values)

for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2,height,int(height),ha='center',va='bottom',fontsize=9)

plt.title('Late Deliveries by Order Hour')
plt.xlabel('Hour of Day')
plt.ylabel('Number of Late Deliveries')
plt.xticks(range(0,24))
plt.grid(axis='y')
plt.show()
```



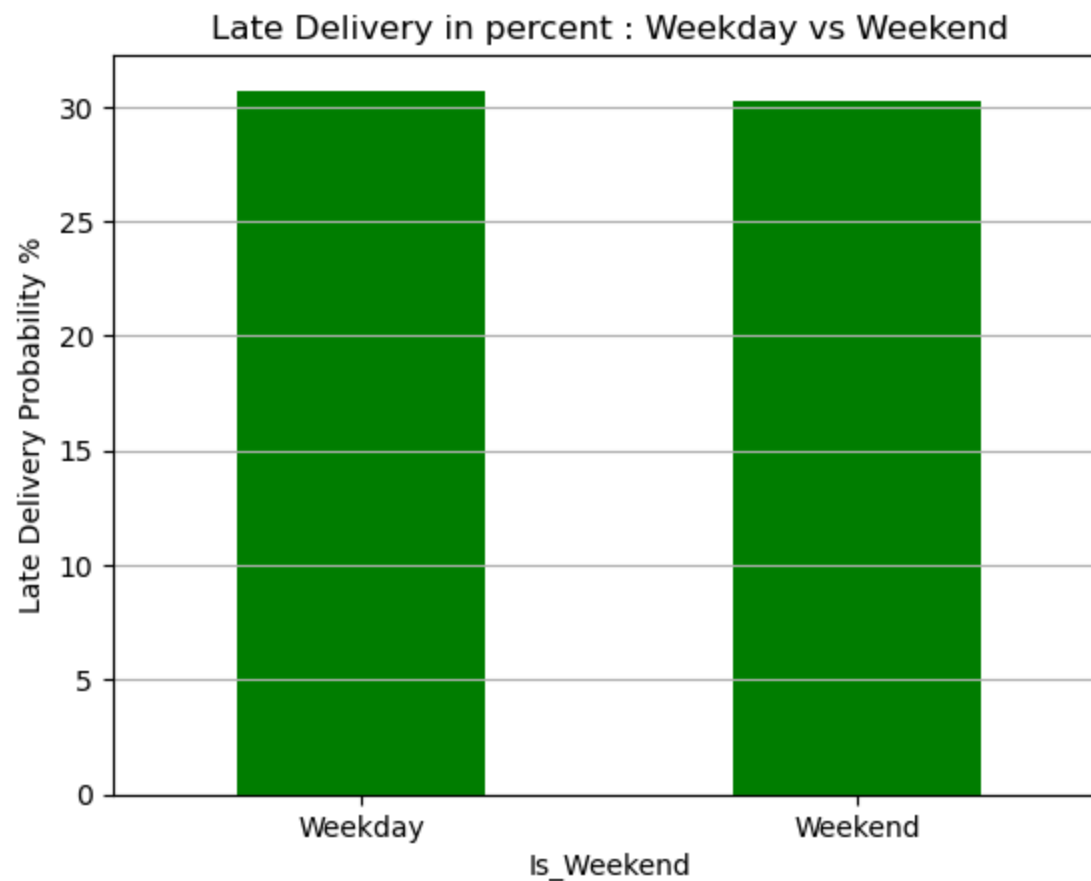
5. Are late Deliveries more common on weekends compared to weekdays??

```
In [145... # Separated Weekdays and Weekends
df['Day_Name'] = df['Order_Date'].dt.day_name()
df['Is_Weekend'] = df['Day_Name'].isin(['Saturday', 'Sunday'])
```

```
In [146... # Comparing Late Delivery Probability
late_weekend = (df.groupby('Is_Weekend')['Late_Delivery'].mean() * 100)
print(late_weekend)
```

```
Is_Weekend
False    30.709087
True     30.260273
Name: Late_Delivery, dtype: float64
```

```
In [147... late_weekend.plot(kind='bar', color = 'g')
plt.title('Late Delivery in percent : Weekday vs Weekend')
plt.ylabel('Late Delivery Probability %')
plt.xticks([0,1], ['Weekday','Weekend'], rotation=0)
plt.grid(axis='y')
plt.show()
```



6. Which Cities or Areas have the highest late delivery Rates??

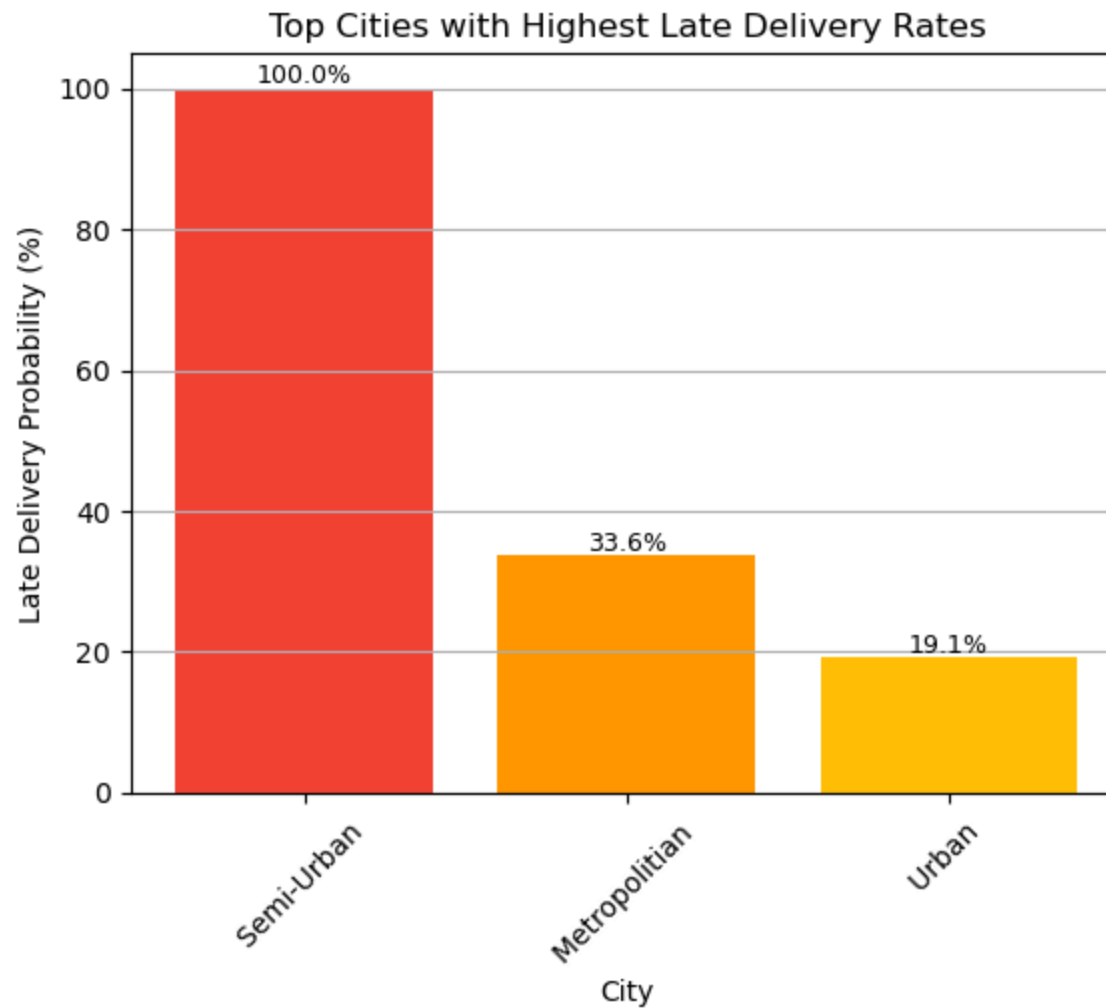
```
In [148... # Late Delivery rate per City
late_by_city = (df.groupby('City')['Late_Delivery'].mean() * 100).sort_values(ascending = False)
print(late_by_city)
```

```
City
Semi-Urban      100.000000
Metropolitian   33.640349
Urban          19.131537
Name: Late_Delivery, dtype: float64
```

```
In [149... bars = plt.bar(late_by_city.index, late_by_city.values, color=['#F44336', '#FF9800', '#FFC107'])

for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, height, f'{height:.1f}%', ha='center', va='bottom', fontsize=9)

plt.title('Top Cities with Highest Late Delivery Rates')
plt.xlabel('City')
plt.ylabel('Late Delivery Probability (%)')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.show()
```



7. What Order Characteristics (Distance, Time, Location) are most associated with late deliveries??

```
In [150... # Distance Vs Late Delivery
distance_delay = (df.groupby('Distance_Band')['Late_Delivery'].mean() * 100).sort_values(ascending = False)
print(distance_delay)
```

```
Distance_Band
10+ km      46.215711
5-10 km     21.551387
2-5 km      13.858175
0-2 km      10.361021
Name: Late_Delivery, dtype: float64
```

```
In [151... # Order Time(Hour) Vs Late Delivery
time_delay = (df.groupby('Order_hour')['Late_Delivery'].mean() * 100).sort_values(ascending = False)
print(time_delay.head(10))
```

```
Order_hour
20      51.750000
21      50.825243
19      49.914237
17      36.604278
13      36.417034
18      35.764944
14      32.650073
12      32.156368
11      30.746791
15      16.052632
Name: Late_Delivery, dtype: float64
```

```
In [152... # Location or City Vs Late Delivery
city_delay = (df.groupby('City')['Late_Delivery'].mean() * 100).sort_values(ascending = False)
print(city_delay)
```

```
City
Semi-Urban      100.000000
Metropolitian    33.640349
Urban           19.131537
Name: Late_Delivery, dtype: float64
```

8. What percentage of late deliveries exceed the SLA by more than 10 minutes

```
In [153... # Percentage of late deliveries where delay > 10 mins beyond SLA
sla_time = 30
severe_buffer = 10
```

```
late_df = df[df['Time_taken(min)'] > sla_time]

#Severe late > 10 mins beyond SLA
severe_late = late_df[late_df['Time_taken(min)'] > (sla_time + severe_buffer)]

severe_percentage = (len(severe_late) / len(late_df)) * 100
print(f"Severe SLA Breach %: {severe_percentage:.2f}%")
```

Severe SLA Breach %: 29.69%

9. Are late deliveries significantly higher during peak hours??

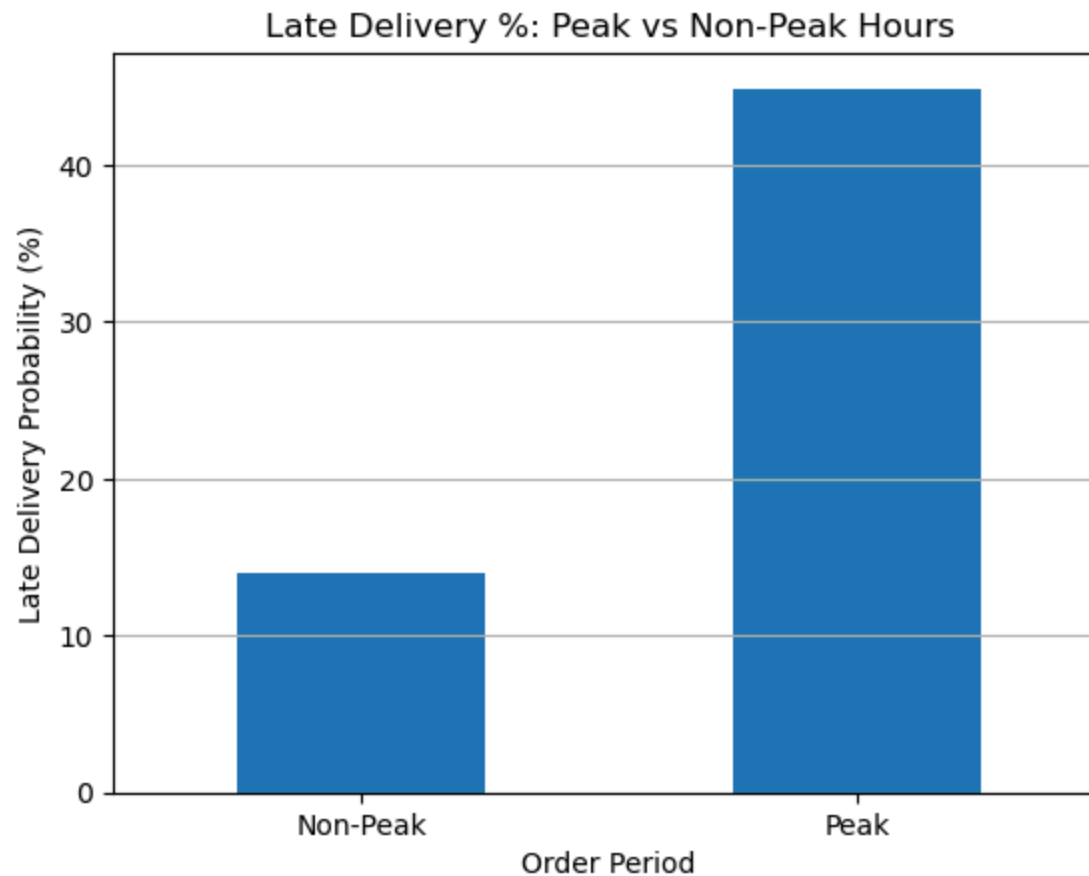
```
In [154... peak_hours = [13, 17, 18, 19, 20, 21]
df['Is_Peak_Hour'] = df['Order_hour'].isin(peak_hours)

# Compare late delivery %
peak_delay = (df.groupby('Is_Peak_Hour')['Late_Delivery'].mean() * 100)
print(peak_delay)
```

```
Is_Peak_Hour
False    13.959147
True     44.871172
Name: Late_Delivery, dtype: float64
```

```
In [155... peak_delay.plot(kind='bar')

plt.title('Late Delivery %: Peak vs Non-Peak Hours')
plt.xlabel('Order Period')
plt.ylabel('Late Delivery Probability (%)')
plt.xticks([0,1], ['Non-Peak', 'Peak'], rotation=0)
plt.grid(axis='y')
plt.show()
```



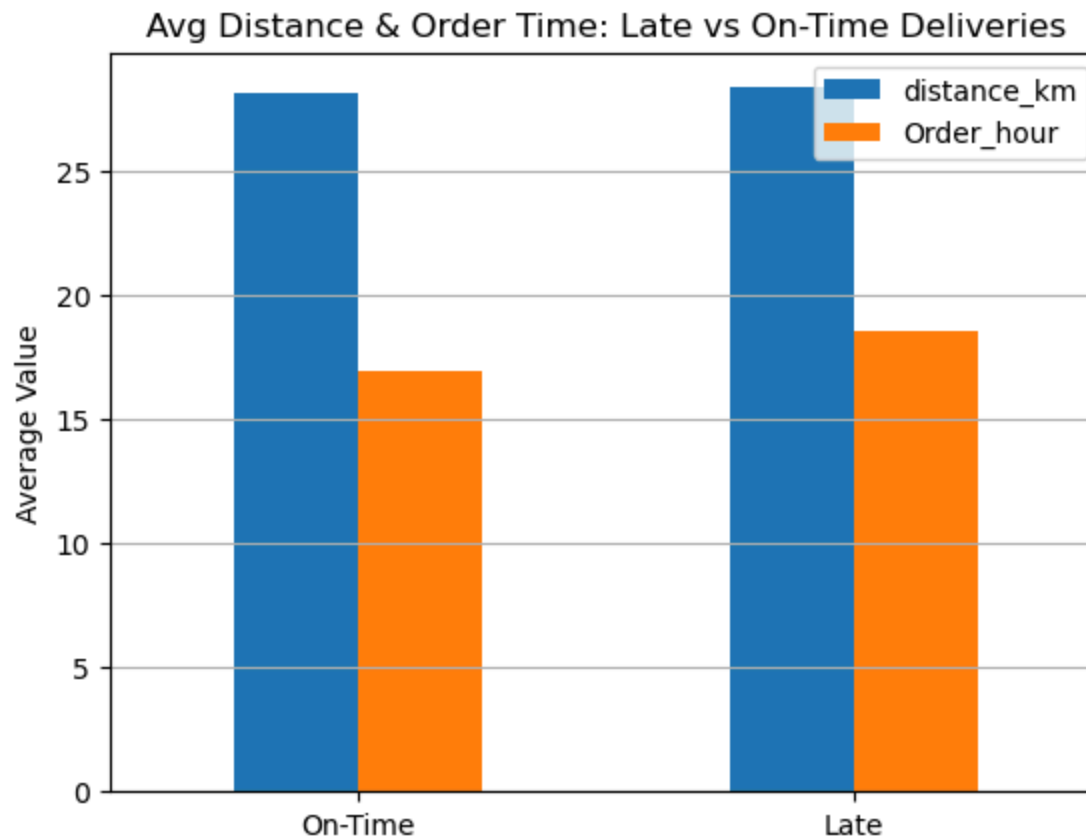
10. How do average distance and order time differ between late and on time deliveries

```
In [156... comparison = (df.groupby('Late_Delivery').agg({'distance_km' : 'mean',  
                                                'Order_hour' : 'mean'}))  
comparison.index = ['On-Time', 'Late']  
print(comparison)
```

	distance_km	Order_hour
On-Time	28.104444	16.936563
Late	28.338947	18.551081


```
In [157... comparison.plot(kind='bar')

plt.title('Avg Distance & Order Time: Late vs On-Time Deliveries')
plt.ylabel('Average Value')
plt.xticks(rotation=0)
plt.grid(axis='y')
plt.show()
```



11. If Delivery Distance is capped at 5 km, how much can late deliveries reduce

```
In [158... # Current Late %
current_late = df['Late_Delivery'].mean() * 100
```

```
# Apply distance cap
capped_df = df[df['distance_km'] <= 5]
# New Late %
capped_late = capped_df['Late_Delivery'].mean() * 100
# Reduction
reduction = current_late - capped_late

print(f"Current Late %: {current_late:.2f}%")
print(f"After 5 km Cap: {capped_late:.2f}%")
print(f"Reduction: {reduction:.2f}%")
```

Current Late %: 30.59%
 After 5 km Cap: 12.69%
 Reduction: 17.90%

In [159...

```
caps = [2, 5, 8, 10]
results = {}

for cap in caps:
    capped_late = df[df['distance_km'] <= cap]['Late_Delivery'].mean() * 100
    results[f'<= {cap} km'] = capped_late

print(results)
```

```
{'<= 2 km': 10.361021426474904, '<= 5 km': 12.688910696761532, '<= 8 km': 16.68526457978505, '<= 10 km': 17.13012826789386}
```

In [160...

```
plt.figure(figsize=(9,5))
bars = plt.bar(results.keys(), results.values(), color = 'r', alpha = 0.6)

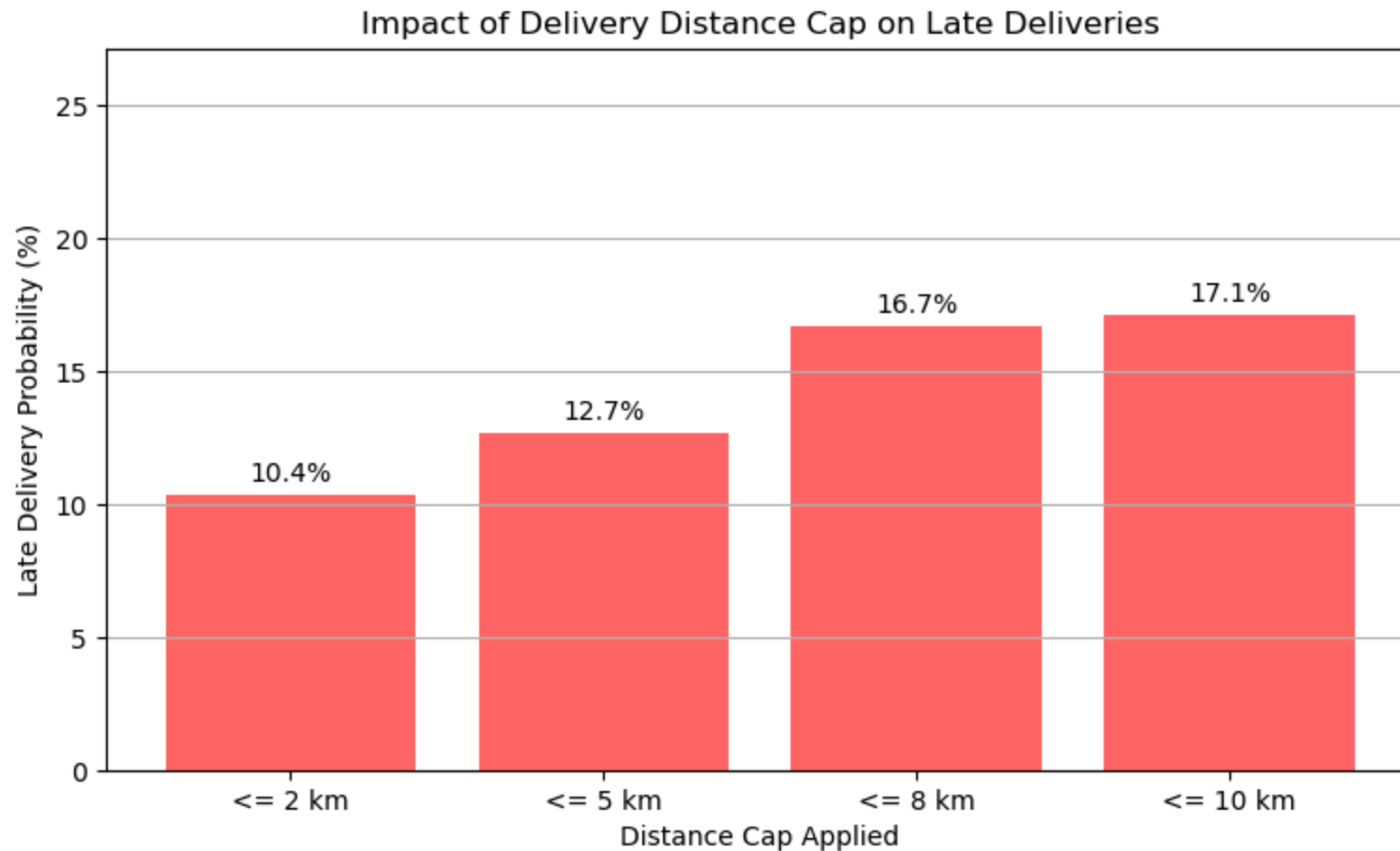
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, height + 0.5, f'{height:.1f}%', ha='center', fontsize=10,)

# Reference Line (current performance)
plt.axhline(current_late, linestyle='--')
plt.text(-0.4, current_late + 0.5, f'Current Late % = {current_late:.1f}%', fontsize=10)

# Titles & Labels
plt.title('Impact of Delivery Distance Cap on Late Deliveries')
plt.xlabel('Distance Cap Applied')
plt.ylabel('Late Delivery Probability (%)')
plt.ylim(0, max(results.values()) + 10)
```

```
plt.grid(axis='y')
plt.show()
```

Current Late % = 30.6%



12. Can we identify high risk orders that are likely to be delivered late

```
In [161]: df['High_Risk_Order'] = ((df['distance_km'] > 8) | (df['Order_hour'].isin([12,13,14,19,20,21])) |
                                   (df['Road_traffic_density'] == 'Jam') | (df['Weatherconditions'].isin(['Stormy', 'Rainy', 'Fog']) |
                                   (df['multiple_deliveries'] > 1) | (df['Delivery_person_Ratings'] < 3.5))
```

```
risk_accuracy = (df.groupby('High_Risk_Order')['Late_Delivery'].mean() * 100)
print(risk_accuracy)

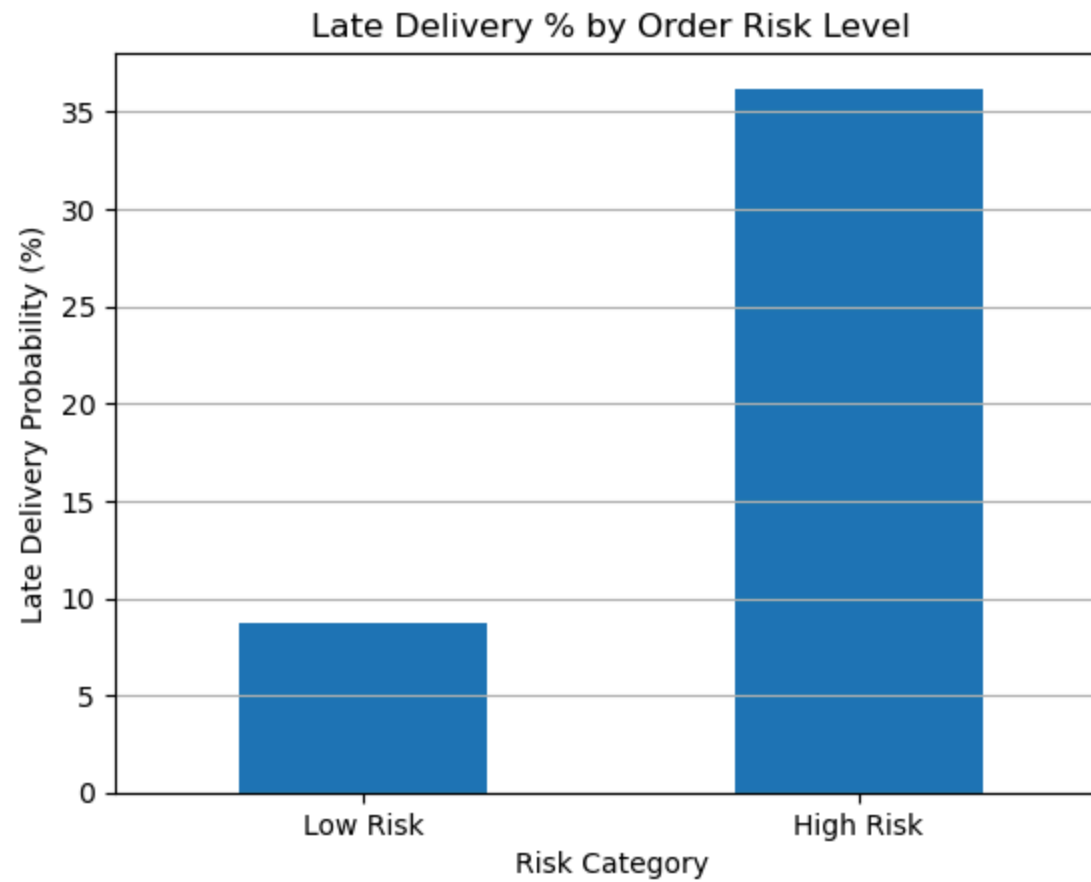
df['High_Risk_Order'].value_counts()
```

```
High_Risk_Order
False      8.717883
True      36.209730
Name: Late_Delivery, dtype: float64
```

```
Out[161...] High_Risk_Order
True      30420
False     7823
Name: count, dtype: int64
```

```
In [162...] risk_accuracy.plot(kind='bar')

plt.title('Late Delivery % by Order Risk Level')
plt.xlabel('Risk Category')
plt.ylabel('Late Delivery Probability (%)')
plt.xticks([0,1], ['Low Risk', 'High Risk'], rotation=0)
plt.grid(axis='y')
plt.show()
```



In []: